Expt No: 02
Date: 23/07/26

Study of Network Cables

Aim:

Study of different types of Network Cables.

a) Understand different types of network Cable.

Different type of cables used in networking are:

1) Unshielded Twisted Pair (UTP) Cable
2) Shielded Twisted Pair (STP) Cable
3) Coaxial Cable
4) Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application use	Image
UTP	Category 3	10 bps	Adv → Cheaper in Cost → Easy to install as they have a smaller overall diameter Disadv → More prone to (EMI) Electromagnetic interference & noise	10Base-T Ethernet Fast Ethernet Gigabit Ethernet Fast Ethernet Gigabit Ethernet	
	Category 5	up to 100Mbps			
	Category 5e	1Gbps			
STP	Category 6, 6a	10Gbps	Adv → Shielded → Faster than UTP → Less susceptible to noise and interference disadv → Expensive → Greater	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centers Gigabit Ethernet 10G Ethernet (100m)	
SSTP	Category 7	10Gbps			

Coaxial Cable	RG-6 RG-59 RG-11	10-100Mbps	Adv →High bandwidth →Immune to interference →Low loss bandwidth →versatile disadv →limited distance →Cost →Size is bulky	speed of Signal is Soom Television network High speed internet Connections	
fibre optics Cable	single mode Multi mode	100 Gbps	Adv →High speed →High bandwidth →High security →Long distance disadv →Expensive →Requires skilled installers	→Maximum distance of fibre optics Cable is around 100 meters	

Result:

The types of network cables and making Cross-wired Cable & straight through cable using clamping / crimping tool